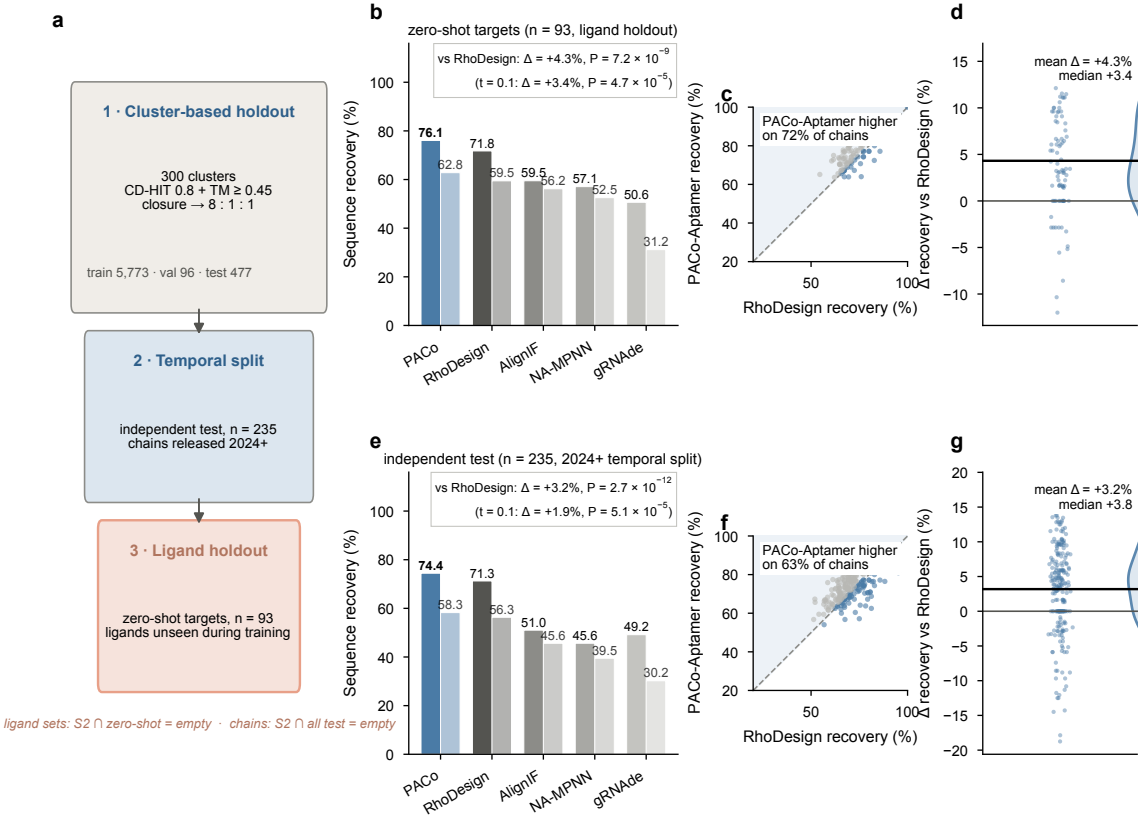

a
1 · Cluster-based holdout
300 clusters
CD-HIT 0.8 + TM ≥ 0.45
closure → 8 : 1 : 1
train 5,773 · val 96 · test 477
2 · Temporal split
independent test, n = 235
chains released 2024+
3 · Ligand holdout
zero-shot targets, n = 93
ligands unseen during training
ligand sets: S2 ∩ zero-shot = empty · chains: S2 ∩ all test = empty
b
zero-shot targets (n = 93, ligand holdout)
vs RhoDesign: Δ = +4.3%, P = 7.2 × 10−9
(t = 0.1: Δ = +3.4%, P = 4.7 × 10−5)
Sequence recovery (%)
76.1
62.8
71.8
59.5
59.5
56.2
57.1
52.5
50.6
31.2
PACo
RhoDesign
AlignIF
NA-MPNN
gRNAde
c
PACo-Aptamer higher on 72% of chains
PACo-Aptamer recovery (%)
RhoDesign recovery (%)
d
mean Δ = +4.3%
median +3.4
Δ recovery vs RhoDesign (%)
e
independent test (n = 235, 2024+ temporal split)
vs RhoDesign: Δ = +3.2%, P = 2.7 × 10−12
(t = 0.1: Δ = +1.9%, P = 5.1 × 10−5)
Sequence recovery (%)
74.4
58.3
71.3
56.3
51.0
45.6
45.6
39.5
49.2
30.2
PACo
RhoDesign
AlignIF
NA-MPNN
gRNAde
f
PACo-Aptamer higher on 63% of chains
PACo-Aptamer recovery (%)
RhoDesign recovery (%)
g
mean Δ = +3.2%
median +3.8
Δ recovery vs RhoDesign (%)